JEE Mains 2019 Chapter wise Question Bank

Ray Optics - Questions

Q1

A convex lens is put 10 cm from a light source and it makes a sharp image on a screen, kept 10 cm from the lens. Now a glass block (refractive index 1.5) of 1.5 cm thickness is placed in contact with the light source. To get the sharp image again, the screem is shifted by a distance d. Then d is:

(1) 1.1 cm away from the lens
(2) 0
(3) 0.55 cm towards the lens
(4) 0.55 cm away from the lens

9 Jan Morning

Q2

Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be:

(1) 45° (2) 60°
(3) 75° (4) 90°

9 Jan Evening

Q3

A plano convex lens of refractive index μ_1 and focal length f_1 is kept in contact with another plano concave lens of refractive index μ_2 and focal length f_2 If the radius of curvature of their spherical faces is R each and $f_1 = 2f_2$, then μ_1 and μ_2 are related as:

(1) $\mu_1 + \mu_2 = 3$ (2) $2\mu_1 - \mu_2 = 1$
(3) $3\mu_2 - 2\mu_1 = 1$ (4) $2\mu_2 - \mu_1 = 1$

10 Jan Morning

Q4

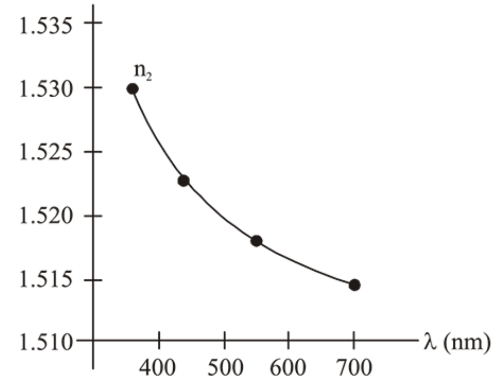
The eye can be regarded as a single refracting surface. The radius of curvature of this surface is equal to that of cornea (7.8 mm). This surface separates two media of refractive indices 1 and 1.34. Calculate the distance from the refracting surface at which a parallel beam of light will come to focus.

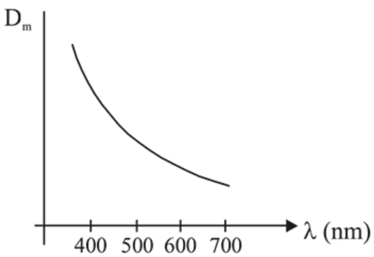
(1) 1 cm (2) 2 cm
(3) 4.0 cm (4) 3.1 cm

10 Jan Evening

Q5

the following graphs is the correct one, if D_m is the angle of minimum deviation ?



(1)


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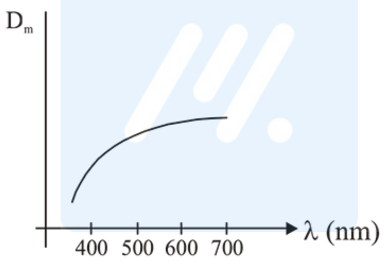
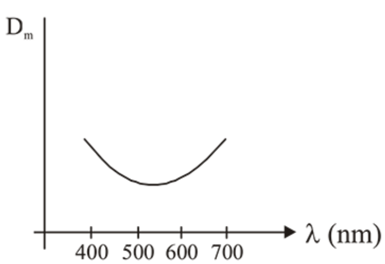


(3)


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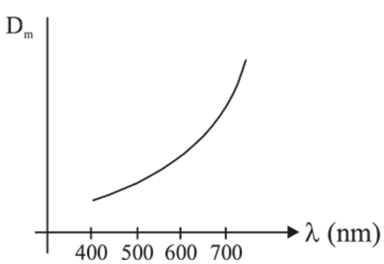

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Q6

An object is at a distance of 20 m from a convex lens of focal length 0.3 m. The lens forms an image of the object. If the object moves away from the lens at a speed of 5m/s, the speed and direction of the image will be :

(1) 2.26×10^{-3} m/s away from the lens
(2) 0.92×10^{-3} m/s away from the lens
(3) 3.22×10^{-3} m/s towards the lens
(4) 1.16×10^{-3} m/s towards the lens

11 Jan Morning

Q7

A monochromatic light is incident at a certain angle on an equilateral triangular prism and suffers minimum deviation. If the refractive index of the material of the prism is $\sqrt{3}$, then the angle of incidence is :

(1) 90° (2) 30°
(3) 60° (4) 45°

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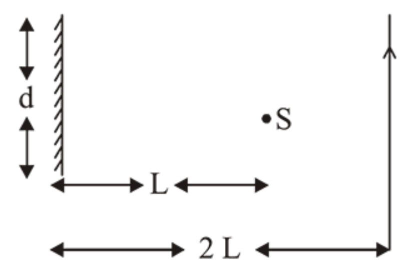
Q8

A point source of light, S is placed at a distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance 2L as shown below. The distance over which the man can see the image of the light source in the mirror is:



(1) d (2) 2d
(3) 3d (4) $\frac{d}{2}$

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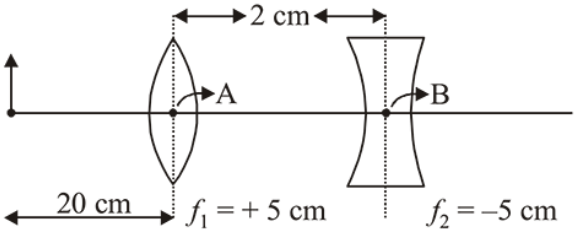
Q9

What is the position and nature of image formed by lens combination shown in figure? (f_1, f_2 are focal lengths)



(1) 70 cm from point B at left; virtual
(2) 40 cm from point B at right; real
(3) $\frac{20}{3}$ cm from point B at right, real
(4) 70 cm from point B at right; real

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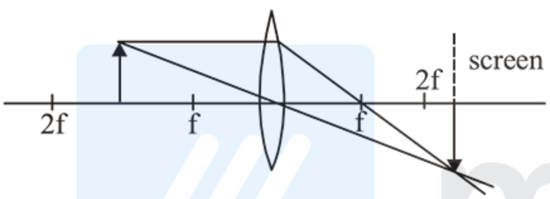
Q10

Formation of real image using a biconvex lens is shown below :



If the whole set up is immersed in water without disturbing the object and the screen positions, what will one observe on the screen ?

(1) Image disappears (2) Magnified image
(3) Erect real image (4) No change

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Q11

A plano-convex lens (focal length f_2, refractive index μ_2, radius of curvature R) fits exactly into a plano-concave lens (focal length f_1, refractive index μ_1, radius of curvature R). Their plane surfaces are parallel to each other. Then, the focal length of the combination will be :

(1) $f_1 - f_2$ (2) $\frac{R}{\mu_2 - \mu_1}$

(3) $\frac{2f_1f_2}{f_1 + f_2}$ (4) $f_1 + f_2$

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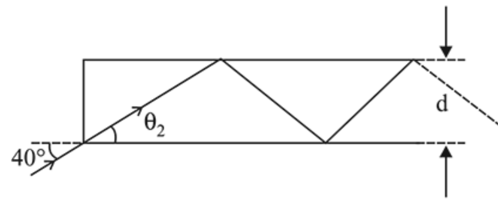
Q12

In figure, the optical fiber is l = 2 m long and has a diameter of d = 20 μm. If a ray of light is incident on one end of the fiber at angle θ_1 = 40°, the number of reflections it makes before emerging from the other end is close to :

(refractive index of fiber is 1.31 and sin 40° = 0.64)



(1) 55000 (2) 66000 (3) 45000 (4) 57000

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Q13

An upright object is placed at a distance of 40 cm in front of a convergent lens of focal length 20 cm. A convergent mirror of focal length 10 cm is placed at a distance of 60 cm on the other side of the lens. The position and size of the final image will be :

(1) 20 cm from the convergent mirror, same size as the object
(2) 40 cm from the convergent mirror, same size as the object
(3) 40 cm from the convergent lens, twice the size of the object
(4) 20 cm from the convergent mirror, twice the size of the object

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Q14

A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance of the object for which this concave mirror, by itself would produce a virtual image would be :

(1) 30 cm (2) 25 cm (3) 10 cm (4) 20 cm

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Q15

A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is:

(1) 0.24 m (2) 1.60 m
(3) 0.32 m (4) 0.16 m

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Q16

A convex lens of focal length 20 cm produces images of the same magnification 2 when an object is kept at two distances x_1 and x_2 ($x_1 > x_2$) from the lens. The ratio of x_1 and x_2 is:

(1) 2 : 1 (2) 3 : 1 (3) 5 : 3 (4) 4 : 3

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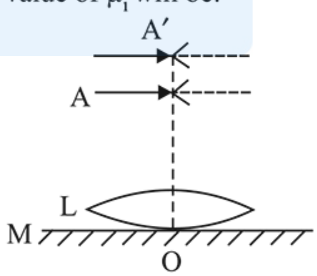
Q17

A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M. When a pin is placed at A, such that OA = 18 cm, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ_i is put between the lens and the mirror, the pin has to be moved to A', such that OA' = 27 cm, to get its inverted real image at A' itself. The value of μ_i will be:



(1) $\frac{4}{3}$ (2) $\frac{3}{2}$ (3) $\sqrt{3}$ (4) $\sqrt{2}$

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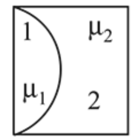
Q18

One plano-convex and one plano-concave lens of same radius of curvature 'R' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2, then the focal length of the combination is :



(1) $\frac{R}{\mu_1 - \mu_2}$ (2) $\frac{2R}{\mu_1 - \mu_2}$

(3) $\frac{2R}{2(\mu_1 - \mu_2)}$ (4) $\frac{R}{2-(\mu_1 - \mu_2)}$

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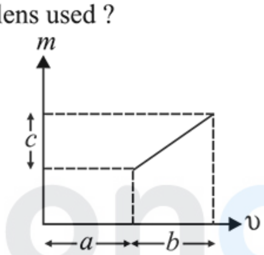
Q19

The graph shows how the magnification m produced by a thin lens varies with image distance *v*. What is the focal length of the lens used ?



(1) $\frac{b^2}{ac}$ (2) $\frac{b^2c}{a}$ (3) $\frac{a}{c}$ (4) $\frac{b}{c}$

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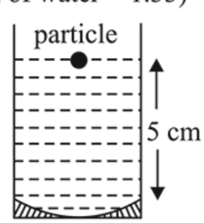
Q20

A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small partricle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to :

(Refractive index of water = 1.33)



(1) 6.7 cm (2) 13.4 cm (3) 8.8 cm (4) 11.7 cm

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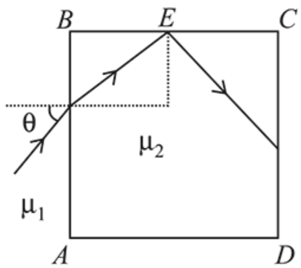
Q21

A transparent cube of side d, made of a material of refractive index μ_2, is immersed in a liquid of refractive index $\mu_1 (\mu_1 < \mu_2)$. A ray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC.



Then θ must satisfy :

(1) $\theta < \sin^{-1}\dfrac{\mu_1}{\mu_2}$ (2) $\theta > \sin^{-1}\sqrt{\dfrac{\mu_2^2}{\mu_1^2}-1}$

(3) $\theta < \sin^{-1}\sqrt{\dfrac{\mu_2^2}{\mu_1^2}-1}$ (4) $\theta > \sin^{-1}\dfrac{\mu_1}{\mu_2}$

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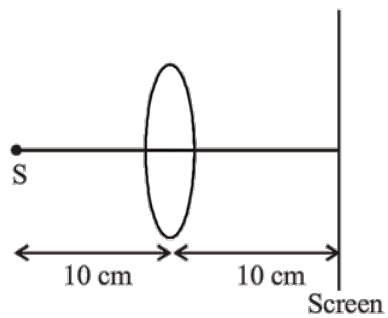
Ray Optics - Answers

Q1

(4)



Using lens formula

$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{10}-\frac{1}{-10}=\frac{1}{f} \Rightarrow f = 5cm$

Shift due to slab, $= t\left(1-\frac{1}{\mu}\right)$ in the direction of incident ray

or, $d = 1.5\left(1-\frac{2}{3}\right) = 0.5$

Now, $u = -9.5$

Again using lens formulas $\frac{1}{v}-\frac{1}{-9.5}=\frac{1}{5}$

$\Rightarrow \frac{1}{v}=\frac{1}{5}-\frac{2}{19}=\frac{9}{95}$

or, $v=\frac{95}{9}=10.55\,cm$

Thus, screen is shifted by a distance $d = 10.55 - 10 = 0.55$ cm away from the lens.

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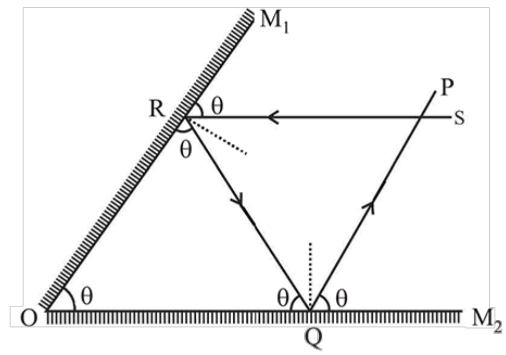
Q2

(2)



Let angle between the two mirrors be θ.

Ray PQ || mirror M_1 and Rs || mirror M_2

$\therefore \quad M_1Rs = \angle ORQ = \angle M_1OM_2 = \theta$

Similarly, $\angle M_2QP = \angle OQR = \angle M_2OM_1 = \theta$

$\therefore \quad$ In ΔORQ, $3\theta = 180° \Rightarrow \theta = \frac{180°}{3} = 60°$

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Q3

(2) From lens maker's formula,

$$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{f_1}=(\mu_1-1)\left(\frac{1}{\infty}-\frac{1}{-R}\right)=\frac{1}{2f_2}$$

Similarly, for plano-concave lens

$$\frac{1}{f_2}=(\mu_2-1)\left(\frac{1}{-R}-\frac{1}{\infty}\right)$$

Dividing $\frac{1}{f_1}$ by $\frac{1}{f_2}$ we get,

$$\frac{(\mu_1-1)}{R}=\frac{(\mu_2-1)}{2R}$$

or, $2\mu_1-\mu_2=1$

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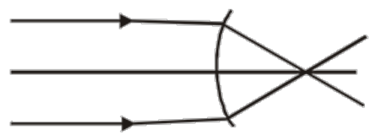
Q4

(4) using, $\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$

$R = 7.8$ mm



$\mu_1 = 1 \ \mu_2 = 1.34$

$\Rightarrow \frac{1.34}{V}-\frac{1}{\infty}=\frac{1.34-1}{7.8}[\because u=\infty]$

$\therefore V = 30.7 \text{ mm} = 3.07 \text{ cm} \simeq 3.1 \text{ cm}$

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Q5

(1) When angle of prism is small, then angle of deviation is given by $D_m = (\mu - 1)A$

So, if wavelength of incident light is increased, μ decreases and hence D_m decreases.

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Q6

(4) By lens formula

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}-\frac{1}{(-20)}=\frac{10}{3}$$

$$\frac{1}{v}=\frac{10}{3}-\frac{1}{20}$$

$$\frac{1}{v}=\frac{197}{60};\ v=\frac{60}{197}$$

Magnification of lens (m) is given by

$$m=\left(\frac{v}{u}\right)=\frac{\left(\frac{60}{197}\right)}{20}$$

velocity of image wrt. to lens is given by

$v_{I/L} = m^2 v_{O/L}$

direction of velocity of image is same as that of object

$v_{O/L} = 5$ m/s

$$v_{I/L}=\left(\frac{60\times 1}{197\times 20}\right)^2(5)$$

$= 1.16 \times 10^{-3}$ m/s towards the lens

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Q7

(3) For minimum deviation:

$$r_1 = r_2 = \frac{A}{2} = 30^o$$

by Snell's law $\mu_1 \sin i = \mu_2 \sin r$

$$1 \times \sin i = \sqrt{3}\times\frac{1}{2}=\frac{\sqrt{3}}{2} \Rightarrow i = 60$$

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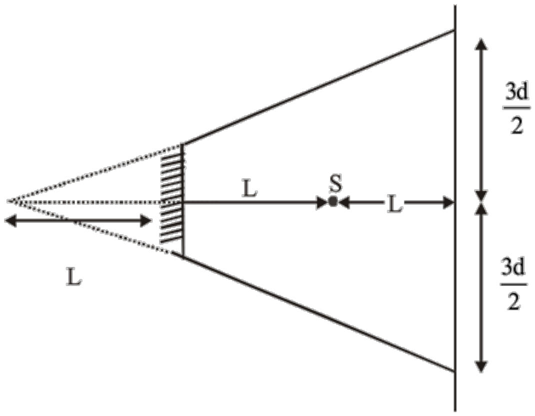
Q8

(3)



Total distance $= \frac{3d}{2} + \frac{3d}{2} = 3d$

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Q9

(4) By lens's formula, $\frac{1}{V} - \frac{1}{u} = \frac{1}{f}$

For first lens, $[u_1 = -20]$

$\frac{1}{V_1} - \frac{1}{-20} = \frac{1}{5} \Rightarrow V_1 = \frac{20}{3}$

Image formed by first lense will behave as an object for second lens

so, $u_2 = \frac{20}{3} - 2 = \frac{14}{3}$

$\frac{1}{V_2} - \frac{1}{\frac{14}{3}} = \frac{1}{-5} \Rightarrow V_2 = 70\,\text{cm}$

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Q10

(1) According to lens maker's formula,

$\frac{1}{f} = (\mu_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Focal length of lens will change due to change in refrective index μ_{rel}. So, image will be formed at new position. Hence image disappears

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Q11

(2) $\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{+1}{R}\right)$

$\frac{1}{f_1} = (\mu_1 - 1)\frac{(-1)}{R}$

Now when combined the focal length is given by

$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

$= (\mu_1 - 1)\frac{(-1)}{R} + (\mu_2 - 1)\frac{+1}{R}$

$= \frac{1}{R}[\mu_2 - \cancel{1} - \mu_1 + \cancel{1}]$

$= \frac{\mu_2 - \mu_1}{R}$

$\Rightarrow \; f = \frac{R}{\mu_2 - \mu_1}$

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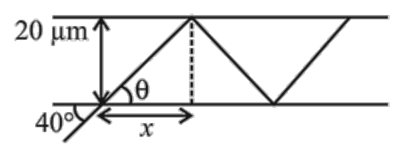
Q12

(4) Using Snell's law of refraction,

$1 \times \sin 40° = 1.31 \sin\theta$

$\Rightarrow \; \sin\theta = \frac{0.64}{1.31} = 0.49 \approx 0.5$

$\Rightarrow \quad \theta = 30°$



$x = 20\ \mu m \times \cot\theta$

$\therefore$ Number of reflections $= \frac{2}{20 \times 10^{-6} \times \cot\theta}$

$= \frac{2 \times 10^6}{20 \times \sqrt{3}} = 57735 \approx 57000$

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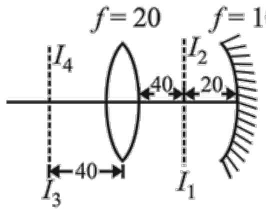
Q13

(Bouns) $v_1 = \dfrac{40 \times 20}{(40-20)} = 40\,cm$



$u_2 = 60 - 40 = 20$ cm

$\therefore v_2 = \dfrac{20 \times 10}{(20-10)} = 20\,cm$

$\therefore$ Image traces back to object itself as image formed by lens is a centre of curvature of mirror.

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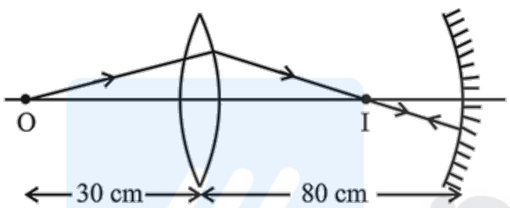
Q14

(3) For lens



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

or $\dfrac{1}{v} - \dfrac{1}{-30} = \dfrac{1}{20}$

$\therefore \quad v = +60$ cm

According to the condition, image formed by lens should be the centre of curvature of the mirror, and so

$2f' = 20$ or $f' = 10$ cm

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Q15

(3) $+5 = -\dfrac{v}{u} \Rightarrow v = -5u$

Using $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f} \Rightarrow \dfrac{1}{-5u} + \dfrac{1}{u} = \dfrac{-1}{0.4}$

$\therefore$ u = – 0.32 m.

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Q16

(2) Using, $M = \dfrac{v}{u}$

or $-2 = \dfrac{v_1}{x_1} \Rightarrow v_1 = -2x_1$

We have $\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$

or $\dfrac{1}{-2x_1} - \dfrac{1}{x_1} = \dfrac{1}{20}$

$x_1 = 30$ cm

And $\dfrac{1}{2x_2} - \dfrac{1}{x_2} = \dfrac{1}{20}$

or $x_2 = -10$ cm

So, $\dfrac{x_1}{x_2} = \dfrac{30}{10} = 3$

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Q17

(1) $\dfrac{1}{f_1} = \dfrac{2}{f_\ell}$

Here $2f_1 = 18$ cm or $f_1 = 9$ cm

So, $\dfrac{1}{9} = \dfrac{2}{f_\ell}$ or $f_\ell = 18$ cm

Using, $\dfrac{1}{f_\ell} = (\mu - 1)\left(\dfrac{2}{R}\right)$

or $\dfrac{1}{18} = (1.5 - 1)\left(\dfrac{2}{R}\right)$

$\therefore R = 18$ cm

when liquid is put between, then

$\dfrac{1}{f_2} = \dfrac{2}{f_\ell} + \dfrac{2}{f}$

or $\dfrac{1}{(27/2)} = \dfrac{2}{18} + \dfrac{2}{f}$

or $f = -54$ cm

Now $-\dfrac{1}{54} = (\mu_1 - 1) \times \dfrac{1}{R}$

$= (\mu_1 - 1) \times \left(\dfrac{1}{-18}\right)$

$\therefore \mu_1 = \dfrac{1}{3} + 1 = \dfrac{4}{3}$

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Q18

(1) Focal length of plano-convex lens-

$$\frac{1}{f_1}=(\mu_1-1)\left(\frac{1}{\infty}-\frac{1}{-R}\right)=\frac{\mu_1-1}{R}$$

$$\Rightarrow f_1=\frac{R}{(\mu_1-1)}$$

Focal length of plano-concave lens -

$$\frac{1}{f_2}=(\mu_2-1)\left(\frac{1}{-R}-\frac{1}{\infty}\right)=\frac{\mu_2-1}{-R}$$

$$\Rightarrow f_2=\frac{-R}{(\mu_2-1)}$$

For the combination of two lens-

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{\mu_1-1}{R}-\frac{\mu_2-1}{R}$$

$$=\frac{\mu_1-\mu_2}{R}$$

$$\Rightarrow f_{eq}=\frac{R}{\mu_1-\mu_2}$$

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Q19

(4) From the equation of line

$m=k_1v+k_2 \qquad (\because y=mx+c)$

$$\Rightarrow \frac{v}{u}=k_1v+k_2 \qquad \left(\because m=\frac{v}{u}\right)$$

$$\Rightarrow \frac{1}{u}=k_1+\frac{k_2}{v}$$

(Dividing both sides by v)

$$\Rightarrow \frac{k_2}{v}-\frac{1}{u}-k_1$$

Comparing with lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$, we get

$k_1=\frac{1}{-f}$ and $k_2=1$

$$\therefore f=\frac{1}{\text{slope of } m-v \text{ graph}}=-\frac{b}{c}$$

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Q20

(3) If v is the distance of image formed by mirror, then

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

or $\frac{1}{v}+\frac{1}{-5}=\frac{1}{-20}$

$$\therefore v=\frac{20}{3}\text{cm}$$

Distance of this image from water surface

$$=\frac{20}{3}+5=\frac{35}{3}\text{cm}$$

Using, $\frac{RD}{AD}=\mu$

$$\therefore AD=d=\frac{RD}{\mu}=\frac{(35/3)}{1.33}=8.8 \text{ cm}$$

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Q21

(3) Using, $\sin\theta_{max}=\mu_1\sqrt{\mu_2^2-\mu_1^2}=\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$

or $\theta_{max}=\sin^{-1}\left(\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}\right)$

For T_1R, $\theta<\sin^{-1}\left(\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}\right)$

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